

Software System Design (2026) - Architecture

Assignment 1

(25 Points)

You are responsible to INDEPENDENTLY complete four individual tasks related to software quality attributes, tactics and design decisions, then compile and submit your report.

DEADLINE: 11:59pm on (Wednesday) May 20, 2026

Task 1: Quality Attribute Scenarios (8 points)

With the rapid evolution of Artificial Intelligence (AI), it has been emerging as an enabling component in the majority of contemporary software systems. Do your own research on **AI-Enabled** and **AI-Native** Applications, discuss your understanding of their **differences**. Figure out **TWO quality attributes** with the highest priority for each type that have to be seriously considered in the architectures of such applications, at least **ONE of each type is NOT introduced** in lectures, and **ONE can be COMMON** between them. Apply scenario-based (stimulus-response) analysis method to create their **general scenarios** and typical **concrete scenarios** (using six-element analysis with at least **THREE response measures**) respectively based on your research. Present the general scenario in the format of six-element table and the concrete scenario in the stimulus-response diagram.

Task 2: Tactics (6 points)

Discuss at least **TWO possible strategies** to improve each quality attribute; propose at least **TWO different tactics** for each possible strategy and identify their potential **impacts on other attributes**, in particular compare the strategies and tactics of the **common quality attribute** between the above two types of applications. Use a table with columns of quality attributes and rows of strategies and tactics to discuss the benefits and the penalties of the potential impacts that may come with each strategy and tactic.

Task 3: Design Decisions (6 points)

Follow the seven categories of architectural design decisions, enumerate at least **TWO** most important **design decisions** (in your point of view) under **each of five categories** (of your choice) respectively for the quality attributes in Task 1. Elaborate the **rationale** supporting each of your proposed design decisions.

Task 4: Alternative Tactics Debate (5 points)

Enumerate the possible benefits and the costs of 'active redundancy', 'passive redundancy' and 'spare' (not covered in Task 2), the alternative tactics introduced for Availability in lecture. Discuss at least **THREE different scenarios** that each of them may better fit in respectively with your reasons behind. Conclude the **criteria** for how to make the right choice among them, and elaborate the **rationale** supporting your argument.

Assignment Deliverable:

The assignment deliverables should be submitted individually. Each deliverable contains no more than 10 pages (A4 size) plus a cover page. Note your individual submission will be plagiarism and AI-generated content checked. You'll receive no point if plagiarism is detected in any task of your assignment.